Heart Disease Risk Modelling

Bayesian Networks and Feed Forward Neural Networks

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Dataset and prediction target

Target: recorded disease presence, num > 0; no future-event prediction

920 source rows → 918 retained; 508 positive and 410 negative

Five numeric and six categorical predictors; hospital and row ID excluded

Training-only imputation, scaling and one-hot encoding; patient linkage uncertain

Validation and leakage controls

Selection S: 550 rows → five common stratified folds; neural seeds 42, 52, 62

Cutoff C: 184 rows → threshold selection after fitting on S

Retrospective T: 184 rows → frozen models and cutoffs only

Hospital holdouts and Cleveland-only CV use development records only

Models and architecture

BN: class-to-feature scaffold plus oldpeak → slope; pgmpy CPT estimation

Naive Bayes: remove the oldpeak → slope edge; keep bins and smoothing

MLP: scaled / encoded inputs → 32 ReLU → 16 ReLU → sigmoid

Dropout 0.2, AdamW decay 0.001, BCE loss and training-only early stopping

Hybrid hypothesis and implementation

Hypothesis: a structured probability summary may complement continuous interactions

Add standardized BN posterior log-odds as one MLP input

Three-fold cross-fitting creates training scores without own-label leakage

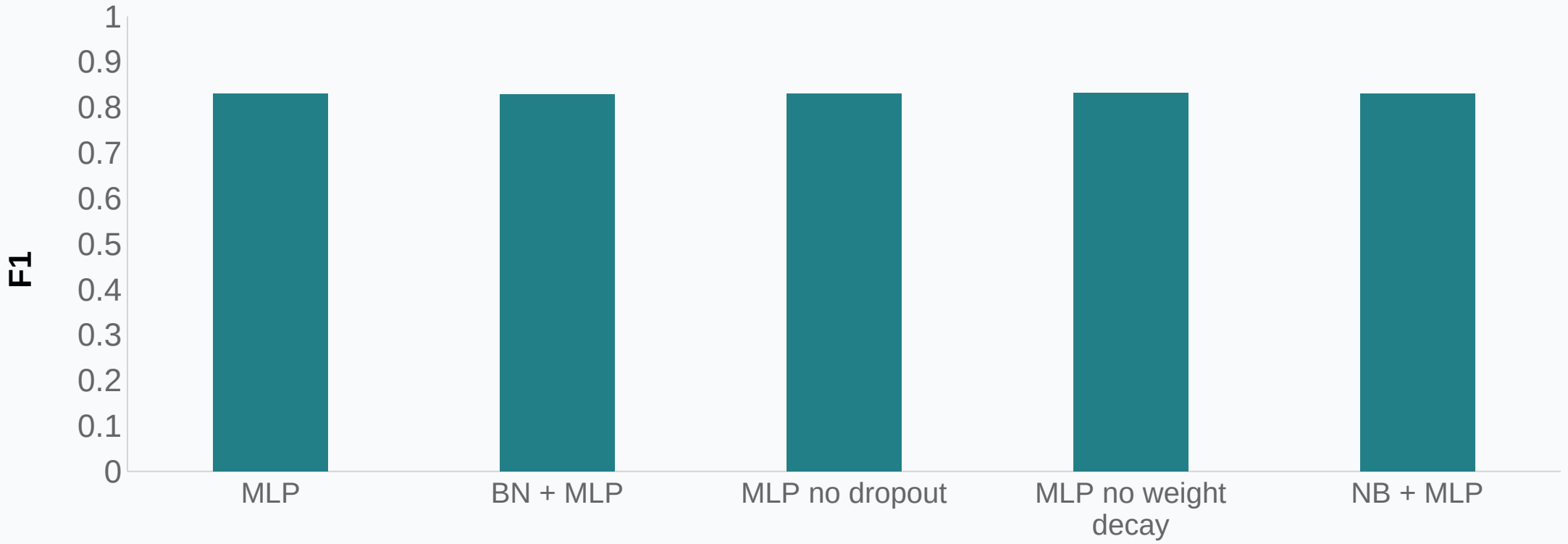
Ablate the BN input; replace BN with NB; remove dropout or weight decay

Primary retrospective comparison

Model	F1	ROC-AUC	PR-AUC	Brier
BN	0.8098	0.8765	0.8936	0.1544
Naive Bayes	0.8103	0.8772	0.8958	0.1532
MLP	0.8411	0.9071	0.9192	0.1236
BN + MLP	0.8358	0.9020	0.9172	0.1252

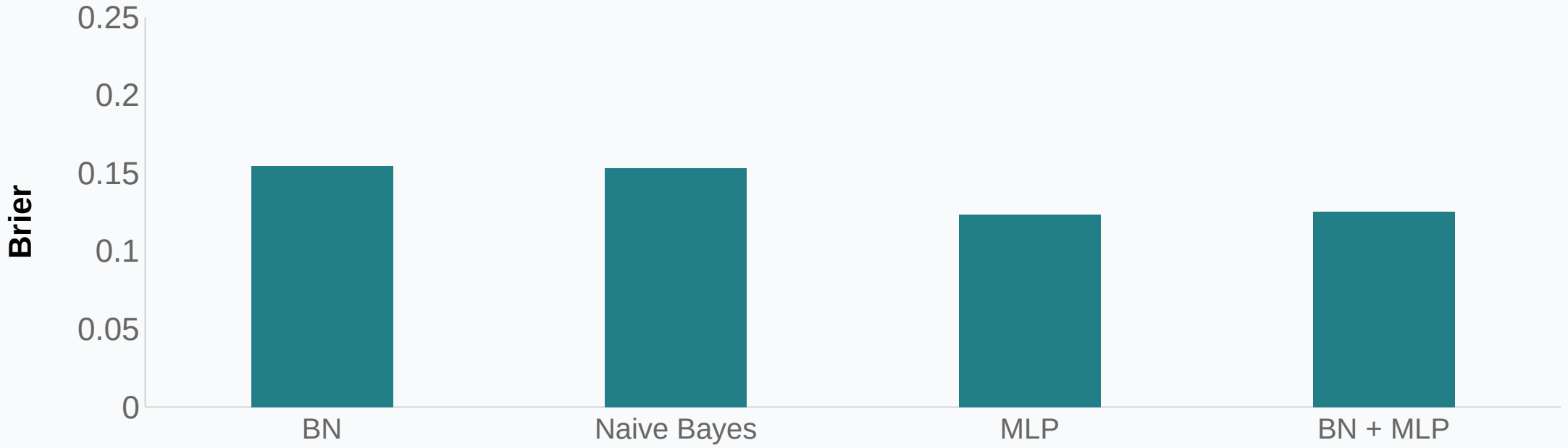
The MLP was selected by S-only CV. Retrospective rankings do not change that decision.

Original feature-hybrid ablations



BN + MLP: 0.8283 | MLP: 0.8296 | MLP without weight decay: 0.8310

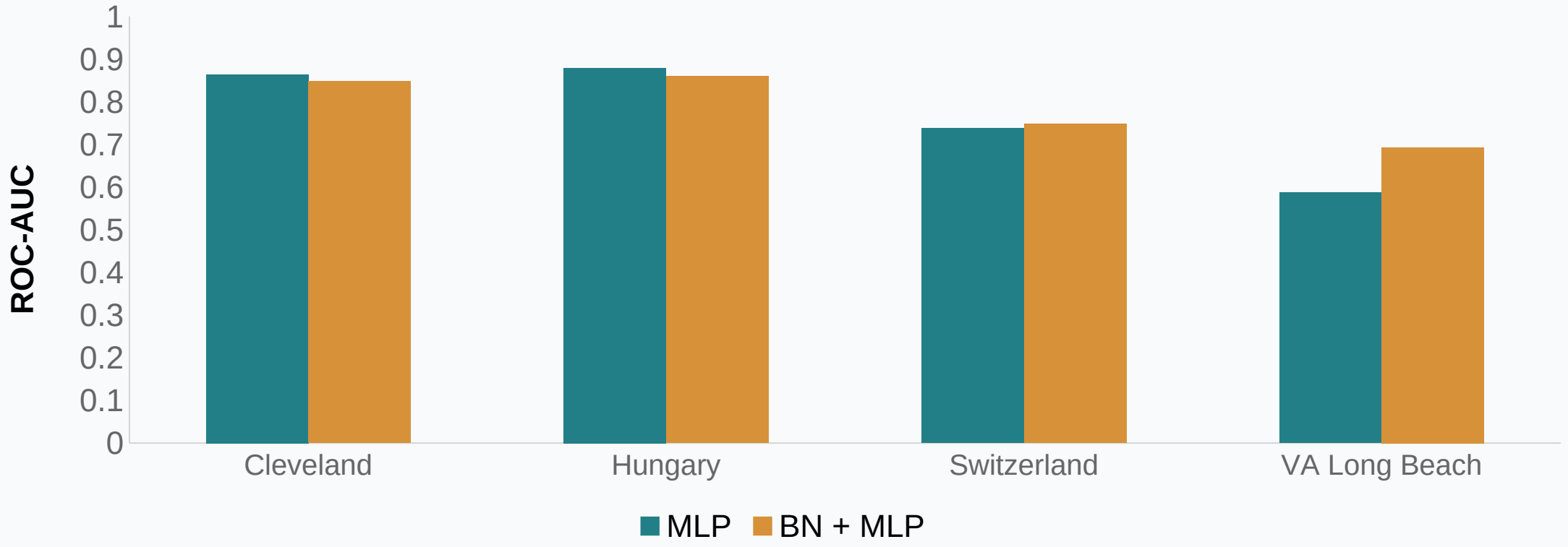
Calibration and operating-point tradeoffs



Hybrid specificity: 0.8171 vs 0.7317; sensitivity: 0.8235 vs 0.8824 for MLP

Five-bin reliability, counts and log loss are saved; clinical calibration is unproven.

Hospital transfer remains uneven



At VA Long Beach, hybrid specificity is only 0.025 despite the AUC gain.

Original experiment conclusions

Hybrid-minus-MLP 95% intervals include zero: F1 −0.0407 to +0.0276

ROC-AUC difference: −0.0157 to +0.0058; Brier: −0.0046 to +0.0078

The hybrid functions, but added value over the MLP is not demonstrated

No prospective validation, causal conclusion or clinical utility is established

Reproducibility and viva preparation

35 original, 105 outer-fold and 12 final models verified

66 local tests and 39 subtests pass, excluding the unchanged Phase 1 UI

Explain cross-fitting, calibration, regularization and the negative hybrid result

Shared review and viva preparation; confirm actual contributions

Follow-up: testing two stronger mechanisms

Residual hybrid: zero initial correction to BN log-odds

Learned log-odds fusion with intercept, rather than a fixed probability average

Controls: tuned MLP, calibrated BN, calibrated MLP and original hybrid

Nested CV: five outer folds, three inner folds, three seeds; C and T excluded

Observed improvement in nested probability scoring

Model	Brier ↓	ROC-AUC	PR-AUC	F1
Calibrated BN-MLP	0.13913	0.87670	0.88437	0.82635
Calibrated BN	0.14038	0.87521	0.88280	0.82276
Tuned MLP	0.14168	0.86896	0.87384	0.82166
Calibrated MLP	0.14228	0.86883	0.87358	0.82174

The calibrated stack has the lowest Brier among all seven tested families.

Observed Brier gains across three seed means

Brier reduction vs tuned MLP: 1.80%

Brier reduction vs calibrated BN: 0.89%

All three seeds improve on both controls; 11/15 seed/fold pairs beat calibrated BN.

Residual Brier: 0.14775, worse than both controls; zero start gives no guarantee.

Follow-up uncertainty and limitations

Conditional 95% interval: -0.00405 to +0.00162; statistical superiority is unestablished

OOF-to-full-fit shift: meta-training and inference score distributions can differ

Post-selection: the best family is chosen and described using the same outer-CV scores

Exploratory S reuse; no untouched external validation or algorithmic novelty claim